

PROJECT PROPOSAL

GROUP PROJECT

CST 394-2

Group No: CST 18

AcademiX

AI-Powered Personalized Learning & Student Performance
Management System

Computer Science and Technology

Department of Computer Science and Informatics

Faculty of Applied Sciences

Uva Wellassa University of Sri Lanka

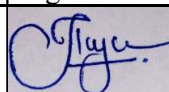
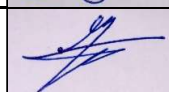
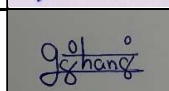
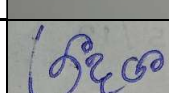
2026

DECLARATION

We hereby declare that the project will be developed by us and will be our own effort and that no part will be plagiarized without citations under the supervision of Ms. D.M.A.Sewwandi, Lecture, Department of Computer Science and Informatics, Faculty of Applied Science, Uva Wellassa University of Sri Lanka. This Project Proposal is submitted for the partial fulfillment of the requirement of the course unit CST 394-2, Project II for the degree of Computer Science and Technology.

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Table of Contents

DECLARATION.....	I
Group Details :	I
Table of Contents	II
1. Introduction.....	4
1.1 Project Title.....	4
1.2 Project Description	4
1.3 Background and Motivation	5
1.4 Problem in Brief	6
1.5 Proposed Solution	7
1.6 Project Aim and Objective	8
1.6.1 Aim	8
1.6.2 Objectives	8
1.7 Novel Features of AcademiX	9
2. Methodology	11
2.1 Agile Methodology	11
2.1.1 Sprint 1: Requirement Analysis & Project Planning.....	11
2.1.2 Sprint 2: System Design & UI/UX Design	12
2.1.3 Sprint 3: Core Development – Lesson, Module & Resource Management	12
2.1.4 Sprint 4: Quiz Management, Study Plan & Attendance Monitoring	13
2.1.5 Sprint 5: ML Integration, Prediction System, Testing & Deployment .	14
2.2 Technical Methodology	15
2.2.1 System Workflow.....	15
2.2.2 Data Collection and Dataset Preparation.....	15
2.2.3 Feature Selection	16
2.2.4 Personalized Study Plan Generation	17
2.2.5 Adaptive Follow-up Quiz Generation	17
2.2.6 Lesson-wise Final Examination Mark Prediction	18
2.2.7 Model Training and Testing	19
2.2.8 Attendance Monitoring and Notification System.....	19
2.2.9 Model Evaluation and Validation.....	19

3. Requirements Identification	21
3.1 Functional Requirements	21
3.2 Non-functional Requirements	25
3.3 System Requirements	27
3.3.1 Hardware Requirements	27
3.3.2 Software Requirements	27
3.3.3 Technologies	27
3.4 User Roles	28
3.4.1 Teacher	28
3.4.2 Student.....	28
4.1 Gantt Chart.....	29
5. References.....	30

List of Figures

Figure 1: The Agile–Scrum Framework	11
Figure 2: Gantt Chart Representing the Project Timeline	29

1. Introduction

1.1 Project Title

AcademiX – AI-Powered Personalized Learning & Student Performance Management System

1.2 Project Description

AcademiX – AI-Powered Personalized Learning & Student Performance Management System is a web-based platform designed to improve teaching effectiveness, monitor student progress, and deliver individualized learning support through intelligent automation. The system supports two user roles. Teacher and Student within a centralized digital learning environment.

Teachers can create module-wise lessons, upload learning resources, and conduct MCQ-based quizzes organized by module under each lesson. The system automatically evaluates quiz submissions, calculates individual student performance percentages, and determines quiz difficulty as Easy, Moderate, or Hard based on the overall class average. These insights help teachers assess both student progress and the effectiveness of their assessments.

Once students complete all module quizzes within a lesson, the system automatically generates personalized study plans tailored to each student's performance level and identified weak areas. Each study plan includes short notes, practice questions, and recommended learning resources focused on weaker modules. Students are given one week to follow their study plan, after which the system generates an adaptive follow-up quiz personalized to their previous performance and weak areas unlike regular module quizzes, which are teacher-created and uniform for all students.

Students can access study materials, attempt quizzes, submit assignments, and monitor their academic progress through the platform. Attendance is recorded via a QR code scanning mechanism, and automated notifications are sent to parents or guardians upon successful attendance marking. A Student Community Hub facilitates peer collaboration, knowledge sharing, and academic discussion, with teachers able to monitor and guide conversations. An Automated Notification System keeps all users informed of quizzes, study plans, assignment deadlines, and other key academic activities.

Additionally, AcademiX integrates a Machine Learning model that predicts each student's expected end examination performance based on their module quiz results and learning patterns. This enables teachers to identify at-risk students early and provide targeted academic support before final examinations, reducing academic risk and improving overall outcomes.

1.3 Background and Motivation

The rapid growth of digital education has transformed traditional learning environments by introducing online platforms, automated assessments, and intelligent academic support systems. However, many existing learning management systems primarily focus on content delivery and basic examination, offering limited capability to track individual student progress, identify weak learning areas, or provide personalized academic guidance [8]. Teachers are often required to manually analyze quiz results, monitor attendance, and support underperforming students, a process that is both time consuming and inconsistent at scale [15].

Research in educational theory highlights that personalized feedback and adaptive learning significantly improve student motivation and academic outcomes [3]. When students receive targeted support based on their individual weaknesses rather than a one-size-fits-all approach, learning effectiveness increases considerably [6]. Adaptive web technologies further support this by enabling systems to respond dynamically to each learner's needs and performance patterns [2]. Collaborative learning environments also strengthen academic engagement by enabling peer-to-peer knowledge sharing and collective problem-solving [11].

Beyond assessment and personalization, the application of educational data mining and machine learning in academic systems has opened new possibilities for predicting student performance and enabling early interventions [8][10]. By analyzing historical quiz results and learning patterns, intelligent systems can estimate students' expected examination outcomes and alert teachers to at-risk learners before final assessments take place [9].

Therefore, AcademiX is developed to address these challenges by integrating lesson management, module-wise MCQ assessments, automated performance analysis, personalized study plan generation, adaptive follow-up quizzes, ML-based

performance prediction, QR-based attendance monitoring, and a student community hub into a single centralized platform eliminating manual workload for teachers while delivering a smarter, more supportive, and data-driven learning experience for every student [13][14].

1.4 Problem in Brief

Many educational institutions still rely on traditional or basic online learning systems that primarily focus on content delivery and standard assessments, without offering intelligent performance tracking or personalized academic support. Teachers frequently face difficulties in manually analyzing quiz results, identifying underperforming students, and preparing individualized support materials, a process that is time-consuming and often leads to delayed academic interventions [8][15].

Students who struggle in specific modules may not receive timely guidance to address their weak areas, causing gaps in understanding to persist over time. Without a structured follow-up mechanism, it becomes difficult to measure whether a student has actually improved after receiving support [6]. Furthermore, traditional systems lack adaptive assessments that adjust to individual student performance, making it impossible to provide truly personalized learning experiences [2]. The absence of data-driven insights also means that teachers have no reliable way to anticipate which students are at risk of underperforming in final examinations until results are already published [8][10].

Beyond academics, attendance monitoring in many institutions remains a manual process, leaving parents uninformed about their child's presence in class in real time. The absence of a collaborative student community within learning platforms also limits peer-to-peer engagement and knowledge sharing [11][12]. Additionally, students and parents often miss critical academic updates due to the lack of an automated notification system, reducing overall engagement with the learning process [13].

Therefore, there is a clear need for an intelligent, all-in-one learning management system that automates quiz evaluation, generates personalized study plans, delivers adaptive follow-up assessments, applies machine learning to predict student examination performance and enable early interventions, monitors attendance with real-time parent notifications, and fosters collaborative learning all within a single integrated platform [14][16].

1.5 Proposed Solution

AcademiX proposes an AI-powered web-based learning and student performance management system that supports Teachers and Students, and within a single centralized platform. Teachers can create lessons organized into modules, upload learning resources, and conduct module-wise MCQ-based quizzes. The system automatically evaluates quiz submissions, calculates each student's performance as a percentage, and determines the overall class average to classify quiz difficulty as Easy, Moderate, or Hard enabling teachers to gain meaningful insights into both student understanding and assessment effectiveness [1][7].

Upon completion of all module quizzes within a lesson, AcademiX automatically generates a personalized study plan for each student based on their identified weak areas. Each plan includes short notes related to weak modules, practice questions tailored to the student's performance level, and recommended study materials and learning resources. After a one-week study period, the system generates adaptive follow-up quizzes personalized to each student's weak areas, allowing continuous and measurable tracking of academic improvement [3][6].

A key advancement in AcademiX is its integrated Machine Learning model, which analyzes each student's module quiz results and learning patterns to predict their expected end examination performance. This predictive capability allows teachers to proactively identify at-risk students and provide targeted revision support, additional academic guidance, and personalized attention well before the final examination takes place enabling timely interventions that reduce academic risk [9][10].

The system also features a QR code-based attendance monitoring mechanism that automatically notifies parents or guardians upon a student's arrival at class, ensuring real-time transparency between the institution and families. A Student Community Hub provides a collaborative space where students can communicate with peers, ask academic questions, and share knowledge, while teachers can monitor discussions and offer guidance [11][12]. An Automated Notification System further keeps all users informed about quiz results, study plans, assignment deadlines, follow-up quizzes, and other key academic activities [13].

Overall, AcademiX eliminates manual workload for teachers, delivers personalized and

data-driven academic support for students, enables predictive performance monitoring, and ensures parents remain actively engaged, all through one intelligent, integrated platform [14][15].

1.6 Project Aim and Objective

1.6.1 Aim

To develop an AI-powered web-based learning and student performance management system that automates quizzes, generates personalized study plans, delivers adaptive follow-up quizzes, predicts student end examination performance using machine learning, monitors attendance with real-time parent notifications, and fosters collaborative learning for Teachers, Students, and Parents within one intelligent, integrated platform.

1.6.2 Objectives

1. To develop a web-based learning management system that allows teachers to manage lessons, modules, learning resources, and conduct module-wise MCQ-based quizzes with automated performance evaluation and difficulty classification.
2. To automatically generate personalized study plans and adaptive follow-up quizzes for each student based on identified weak modules, including short notes, practice questions, and recommended learning resources.
3. To implement a machine learning model that predicts each student's expected end examination performance based on module quiz results and learning patterns, enabling teachers to identify at-risk students and provide timely academic interventions.
4. To implement a QR code-based attendance monitoring system with real-time parent notifications, a Student Community Hub for peer collaboration and teacher-guided discussions, and an Automated Notification System to keep all users informed of key academic activities.

1.7 Novel Features of AcademiX

AcademiX introduces several distinctive features that differentiate it from conventional learning management systems:

1. Automatic Quiz Difficulty Classification

Unlike standard systems that treat all quizzes equally, AcademiX statistically determines quiz difficulty as Easy, Moderate, or Hard based on the calculated class average performance, providing teachers with meaningful insight into both student understanding and assessment effectiveness.

2. AI-Generated Personalized Study Plans

Upon completing all module quizzes within a lesson, the system automatically generates a unique study plan for each individual student based on their specific weak modules, including short notes, practice questions, and recommended learning resources rather than providing generic support materials to all students.

3. Adaptive Follow-Up Quizzes

After the one-week study plan period, the system generates personalized follow-up quizzes tailored to each student's previously identified weak areas. Unlike regular module quizzes which are uniform for all students, these quizzes adapt to individual performance history, enabling continuous and measurable academic improvement.

4. ML-Based Examination Performance Prediction

AcademiX integrates a machine learning model that analyzes each student's module quiz results and learning patterns to predict their expected end examination performance. This allows teachers to proactively identify at-risk students and deliver targeted academic support before final examinations take place, a capability absent in most conventional learning platforms.

5. QR Code Attendance with Real-Time Parent Notification

Student attendance is recorded through QR code scanning, and parents or guardians receive an automated notification immediately upon their child's arrival at class, ensuring real-time transparency between the institution and families.

6. Integrated Student Community Hub

AcademiX embeds a collaborative learning community directly within the platform, allowing students to communicate, share knowledge, and seek peer support, while teachers can monitor discussions and provide academic guidance — all without leaving the system.

2. Methodology

2.1 Agile Methodology

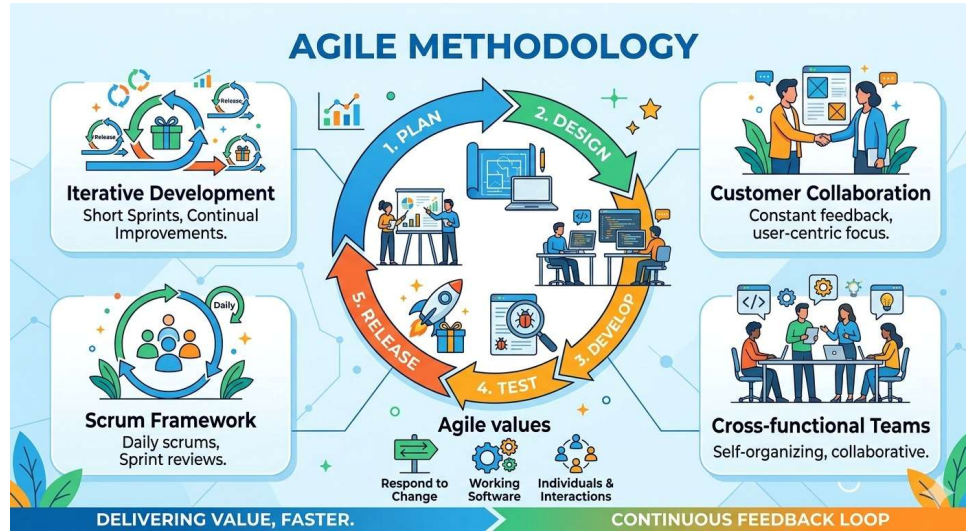


Figure 1: The Agile–Scrum Framework

The development of AcademiX follows the Agile Software Development methodology using the Scrum framework [16]. Agile Scrum is selected because it supports iterative development, continuous feedback collection, flexible requirement handling, and incremental system delivery throughout the project lifecycle [15]. Since the system contains multiple integrated functionalities such as lesson management, quiz automation, personalized learning, attendance monitoring, adaptive follow-up quizzes, student collaboration, and Machine Learning-based examination prediction, Agile Scrum helps divide the project into manageable sprints and ensures continuous improvement during development.

The overall methodology of AcademiX combines software engineering practices [15], educational data analysis [8][10], adaptive learning techniques [2][3], and Machine Learning technologies to improve teaching effectiveness and student academic performance.

2.1.1 Sprint 1: Requirement Analysis & Project Planning

This sprint focuses on gathering system requirements, identifying user roles, analyzing learning workflows, and preparing the project plan.

Main activities include:

- Identifying system users such as teachers and students
- Analyzing lesson and module management workflows
- Understanding quiz and assessment processes
- Identifying personalized learning requirements
- Planning attendance monitoring and notification features
- Defining Machine Learning prediction objectives
- Preparing project schedules and sprint backlogs

During this phase, both functional and non-functional requirements are identified and documented.

2.1.2 Sprint 2: System Design & UI/UX Design

This sprint focuses on designing the system architecture, database structure, frontend and backend workflows, and user interfaces.

Main activities include:

- Designing the database schema
- Designing frontend interfaces for teachers and students
- Planning backend architecture and API structure
- Designing dashboards and performance reporting interfaces
- Designing adaptive quiz workflows
- Designing ML integration workflows
- Creating responsive UI/UX designs

The system architecture is designed to support scalability, modularity, and future AI feature enhancements.

2.1.3 Sprint 3: Core Development – Lesson, Module & Resource Management

This sprint focuses on implementing the core functionalities of the AcademiX platform.

Main development activities include:

- User authentication and authorization
- Lesson and module management
- Learning resource uploading and management
- Assignment management

- Student dashboard implementation
- Teacher dashboard implementation
- Student Community Hub development
- Notification management functionalities

The Student Community Hub is implemented as a collaborative learning environment where students can discuss academic topics, ask questions, and share learning resources.

2.1.4 Sprint 4: Quiz Management, Study Plan & Attendance Monitoring

This sprint focuses on implementing assessment, personalized learning, and attendance monitoring functionalities.

Main activities include:

- MCQ quiz management system development
- Automatic quiz evaluation
- Quiz score calculation
- Quiz difficulty analysis
- Personalized study plan generation
- Adaptive follow-up quiz generation
- QR code-based attendance marking
- Attendance monitoring
- Parent notification automation
- Student performance reporting

After students complete lesson-related module quizzes, the system analyzes their performance and automatically generates personalized study plans according to identified weak learning areas.

Quiz difficulty levels are determined using overall class performance averages. Based on the average quiz score, quizzes are categorized as Easy, Moderate, or Hard.

2.1.5 Sprint 5: ML Integration, Prediction System, Testing & Deployment

This sprint focuses on integrating Machine Learning functionalities, lesson-wise examination prediction, testing, and deployment preparation.

Main activities include:

- Machine Learning model integration
- Lesson-wise examination mark prediction
- Weak lesson area detection
- Follow-up quiz performance analysis
- Prediction result visualization
- System testing and debugging
- Deployment preparation

The Machine Learning model predicts students' expected lesson-wise final examination marks using module quiz marks and adaptive follow-up quiz performance after personalized learning activities are completed.

Continuous Agile Activities (All Sprints)

The following Agile activities are continuously performed throughout all sprints:

- Sprint reviews and stakeholder feedback collection
- Continuous testing and debugging
- Progress monitoring and sprint planning
- Documentation updates
- Task management and version control
- Feature refinement and improvement

2.2 Technical Methodology

2.2.1 System Workflow

The AcademiX system follows a sequential personalized learning and prediction workflow to analyze student performance and estimate lesson-wise final examination marks.

The workflow of the system is as follows:

1. Teachers create lessons and module-wise MCQ quizzes
2. Students complete module quizzes related to each lesson
3. The system automatically evaluates quiz answers and calculates performance percentages
4. Weak learning areas are identified based on quiz performance
5. Personalized study plans are generated according to weak modules
6. Students complete the generated study plans within one week
7. Adaptive follow-up quizzes are generated according to weak areas and student performance
8. Students complete the follow-up quizzes
9. The Machine Learning model analyzes module quiz marks and follow-up quiz performance
10. The system predicts lesson-wise final examination marks
11. Teachers analyze prediction results and provide additional academic support for weak students

This workflow enables continuous learning improvement and early academic intervention before final examinations.

2.2.2 Data Collection and Dataset Preparation

The system uses historical and ongoing quiz performance data collected through the AcademiX platform and previously conducted ICT tuition class assessments. The collected dataset is used to train and evaluate the Machine Learning model responsible for predicting students' lesson-wise final examination marks.

The dataset mainly focuses on module quiz performance and adaptive follow-up quiz results related to each lesson.

The dataset includes:

- Student ID
- Lesson name
- Module-wise quiz marks
- Follow-up quiz marks
- Quiz difficulty level
- Lesson average performance
- Quiz attendance records
- Student performance percentages
- Actual final examination lesson-wise marks

Dataset preparation includes:

- Removing incomplete or duplicated records
- Handling missing values
- Organizing lesson and module information properly
- Assigning numerical values for quiz difficulty levels
- Normalizing data when necessary
- Converting data into structured datasets suitable for Machine Learning

The prepared dataset is divided into training and testing datasets before model development.

2.2.3 Feature Selection

Feature selection is performed to identify the most important attributes affecting students' lesson-wise final examination performance.

The main selected features are:

- Module-wise quiz marks
- Follow-up quiz marks
- Quiz difficulty level
- Lesson average performance
- Student improvement percentage
- Quiz attendance records
- Weak module count

Module quiz marks and follow-up quiz marks are considered the most important

features because they represent both the student's initial understanding and the improvement achieved after personalized learning activities.

Quiz difficulty level is also considered important because different quizzes contain different complexity levels.

2.2.4 Personalized Study Plan Generation

After students complete lesson-related module quizzes, the system automatically identifies weak learning areas according to quiz performance and generates personalized study plans.

The study plan includes:

- Short notes related to weak modules
- Practice questions based on performance level
- Recommended learning materials and study resources
- Personalized revision activities

Students are provided a one-week learning period to complete the personalized study plan before attempting adaptive follow-up quizzes.

2.2.5 Adaptive Follow-up Quiz Generation

After the study plan completion period, the system automatically generates adaptive follow-up quizzes according to each student's weak learning areas and previous performance.

Unlike regular module quizzes, which use teacher-added questions and are the same for all students, follow-up quizzes are adaptive and personalized.

The system generates follow-up quizzes according to:

- Weak learning modules
- Student performance percentages
- Previous quiz mistakes
- Difficulty levels
- Personalized study plan progress

These quizzes are designed to reinforce weak learning areas and measure student improvement after personalized learning activities.

2.2.6 Lesson-wise Final Examination Mark Prediction

After students complete both module quizzes and adaptive follow-up quizzes, the system uses a Machine Learning Regression model to predict students' lesson-wise marks in the final examination.

The system uses the Random Forest Regression or XGBoost Regression algorithm to perform the prediction process.

The prediction model analyzes:

- Module-wise quiz marks
- Follow-up quiz marks
- Quiz difficulty levels
- Lesson average performance
- Student improvement trends

Using these features, the model predicts how many marks a student is likely to obtain from each lesson-related section in the final examination paper.

For example, if a lesson contains multiple modules and students complete both module quizzes and adaptive follow-up quizzes, the model estimates the expected marks for that lesson in the final examination.

This prediction helps teachers:

- Analyze student performance before examinations
- Detect weak lesson areas
- Provide additional academic support
- Conduct early interventions
- Improve student examination readiness

Random Forest Regression or XGBoost Regression is selected because:

- It provides reliable prediction accuracy
- It handles structured educational datasets effectively
- It reduces overfitting problems
- It performs well with multiple performance-related features
- It identifies hidden relationships within educational data

2.2.7 Model Training and Testing

The model training process includes:

- Collecting previous quiz and examination records
- Preparing and preprocessing datasets
- Performing feature engineering
- Splitting datasets into training and testing datasets
- Training the Random Forest Regression or XGBoost Regression model using Python Scikit-learn

The dataset is divided into:

- 80% Training Data
- 20% Testing Data

During training, the model learns relationships between module quiz performance, follow-up quiz performance, and actual lesson-wise final examination marks.

The trained model is tested using separate testing datasets by comparing predicted marks with actual examination results.

2.2.8 Attendance Monitoring and Notification System

The system includes a QR code-based attendance marking mechanism to improve attendance monitoring efficiency.

Main functionalities include:

- QR code scanning for attendance marking
- Real-time attendance recording
- Absentee identification
- Automated parent notifications
- Attendance history management

Once attendance is marked successfully, the system automatically sends notifications to parents or guardians informing them about student attendance status.

2.2.9 Model Evaluation and Validation

The performance of the Random Forest Regression or XGBoost Regression model is evaluated using standard regression evaluation metrics.

The evaluation metrics include:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Model validation is conducted by comparing predicted lesson-wise final examination marks with actual examination results.

The effectiveness of personalized study plans is evaluated using:

- Follow-up quiz performance
- Student improvement percentages
- Weak area improvement analysis
- Comparison between predicted and actual examination performance

These evaluations help measure both the prediction accuracy of the model and the effectiveness of the personalized learning support provided by the system.

3. Requirements Identification

3.1 Functional Requirements

1. User Login and Authentication

- Teacher and Students can securely log in to the system.
- The system validates usernames and passwords during login.
- Users are redirected to their respective dashboards after authentication.
- Invalid login credentials must prevent unauthorized access.
- Students can update their profile information.

2. Lesson and Module Management

- Teacher can add, update, and manage lessons.
- Each lesson can contain multiple modules.
- Teacher can organize modules according to subjects or topics.
- Students can view available lessons and modules.

3. Learning Resource Management

- Teacher can upload learning resources related to each module.
- Resources may include documents, notes, PDFs, and reference links.
- Students can access and download learning resources.
- The system stores uploaded resources securely for future access.
- Resources uploaded by teachers are also utilized by the AI system to generate personalized short notes within study plans.

4. Quiz Management System

- Teacher can add module-wise MCQ quizzes.
- Teacher can add, edit, and manage question sets and answers.
- The system randomly shuffles questions for each student from the predefined question set.
- Students can attend quizzes online within the system.
- The system automatically evaluates quiz answers and calculates

individual student scores as a percentage.

- The system calculates the class average and classifies quiz difficulty as Easy, Moderate, or Hard based on defined thresholds.
- Quiz results are stored in the database for future analysis.

5. Student Performance Analysis

- The system calculates individual student performance percentages for each module quiz.
- Teachers can view performance reports and quiz difficulty classifications for each quiz.
- Students receive notifications regarding their academic performance results.
- Performance data is stored and used for generating personalized study plans, adaptive follow-up quizzes, and end exam predictions.

6. Student Attendance Monitoring

- Student attendance is recorded using a QR code scanning mechanism integrated within the system.
- Upon successful attendance marking, the system automatically sends a real-time notification to the student's parent or guardian confirming their child's arrival.
- Teachers can view attendance records and monitor individual student presence over time.
- The system maintains a complete attendance history for each student for reporting and monitoring purposes.

7. Personalized Study Plan Generation

- The system automatically generates personalized study plans for each student upon completion of all module quizzes within a lesson.
- Study plans are created based on individually identified weak modules according to their performance percentages.
- Short notes are AI-generated from weak module content and teacher-

uploaded resources.

- Practice questions are included based on the student's performance level.
- Recommended learning resources and study materials are suggested for each weak module.
- Students are given a one-week period to follow their personalized study plan.

8. Adaptive Follow-Up Quiz

- The system automatically generates a personalized follow-up quiz for each student after the one-week study plan period.
- Follow-up quizzes are adaptive and tailored to each student's previously identified weak modules and performance history.
- Unlike regular module quizzes, follow-up quizzes are unique for each student.
- Follow-up quiz results are stored to measure academic improvement over time.

9. End Examination Performance Prediction

- The system integrates a Machine Learning model to predict each student's expected end examination performance.
- The ML model analyzes historical module quiz results and individual learning patterns to generate performance predictions.
- Teachers can view predicted end exam scores for each student through the system dashboard.
- Students identified as being at risk of lower end exam performance are flagged for early academic intervention.
- Teachers can use prediction insights to provide additional support, revision guidance, and personalized attention to at-risk students before the final examination.

10.Student Community Hub

- Students can post academic questions, share knowledge, and engage in discussions within the community hub.
- Students can communicate and collaborate with peers on academic topics.
- Teachers can monitor community discussions and provide guidance when necessary.
- The system moderates community content to maintain a productive academic environment.

11.Automated Notification System

- The system sends automated notifications to students and parents regarding attendance, quiz results, study plans, follow-up quizzes, and assignment deadlines.
- Notifications are delivered in real time to ensure timely awareness of academic activities.
- Teachers receive relevant notifications about class performance and student academic updates.

3.2 Non-functional Requirements

1. Performance

- The system should process AI-generated study plans, adaptive follow-up quizzes, and ML-based end exam predictions within an acceptable response time.
- The system should support multiple users simultaneously without significant delays.

2. Security

- User passwords must be securely stored using encryption techniques.
- Only authorized users can access system functionalities based on their roles.
- Student records, quiz data, and attendance information must be protected from unauthorized access.

3. Usability

- The system should provide a clean, intuitive interface for Teachers and Students.
- Navigation across lessons, quizzes, study plans, attendance, and community features should be simple and consistent.

4. Reliability

- The system should ensure accurate quiz evaluation, performance calculation, and data storage at all times.
- The ML-based prediction model must produce consistent and accurate results based on available student performance data.
- QR code attendance marking and automated notifications must function consistently without failure.

5. Scalability

- The system should support future expansion with additional users, lessons, quizzes, and new features without major structural changes.

6. Availability

- The system should be accessible to users at any time through the web platform.
- The platform should minimize downtime during operation.

7. Maintainability

- The system should be easy to update and maintain.
- Developers should be able to modify modules and features with minimal complexity.

3.3 System Requirements

3.3.1 Hardware Requirements

- Router with an Internet connection
- Laptop / Desktop computer with the following requirements
 - Hard Disk - 10 GB or higher
 - RAM - 8 GB or higher
 - Operating System - Windows 10/11, MacOS
 - Processor – i5 or higher
- Storage Device to store backups, project files, and other related data.

3.3.2 Software Requirements

- **IDE** - Visual Studio Code
- **Package Manager** – npm
- **Database Management Tool** - MongoDB Compass
- **API Testing Tool** – Postman
- **Browser** - Google Chrome
- **Diagram Tools** - draw.io
- **User Interface Design Tool** - Figma

3.3.3 Technologies

- **Front End** – Tailwind CSS, React.js
- **Back End** – Node.js, Express.js
- **Database** – MongoDB
- **Machine Learning** - Python, Scikit-learn
- **API Communication** - REST API, Flask API
- **Version Control** - Git, GitHub
- **Development Stack** - MERN Stack (MongoDB, Express.js, React.js, Node.js)

3.4 User Roles

3.4.1 Teacher

- Securely log in to the system
- Add, update, and manage lessons and modules
- Upload learning resources related to each module
- Create and manage module-wise MCQ quizzes and question sets
- View automatically evaluated quiz results, student scores, and quiz difficulty classifications
- View student performance reports and monitor weak students per module
- View and monitor personalized study plans generated for students
- Monitor follow-up quiz results and track student improvement progress
- View AI-based predicted end examination performance of students
- Identify academically at-risk students using predictive analytics
- View student attendance records
- Monitor Student Community Hub discussions and provide academic guidance
- Receive automated notifications regarding class performance and academic updates

3.4.2 Student

- Securely log in to the system
- Access lessons, modules, and uploaded learning resources
- Attend module-wise online MCQ quizzes with shuffled questions
- View quiz scores and individual academic performance records
- Access personalized study plans including short notes, practice questions, and recommended resources
- Attend adaptive follow-up quizzes after the study plan period
- Participate in the Student Community Hub to collaborate and share knowledge with peers
- Receive automated notifications regarding quiz results, study plans, follow-up quizzes, and assignment deadlines
- Update personal profile information

4. Project Plan

4.1 Gantt Chart

Task	Week														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Client Interaction & Requirement Refinement															
Initial Planning & System Setup															
UI/UX Design (Iterative)															
Core Development (Modules)															
Continuous Testing & Debugging															
Feature Enhancements (Based on Feedback)															
Integration & System Testing															
Deployment Preparation															
Final Deployment & Maintenance															
Documentation (Continuous)															

Figure 2: Gantt Chart Representing the Project Timeline

5. References

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